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Fiber Optic Diagnostics: A Breakthrough in Sustained Annulus Pressure Management

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Abstract

This work explores the application of optical fiber-based sensing technologies to monitor and characterize sustained annular pressure behavior in wellbores. Identifying and mitigating this sustained pressure in annuli is critical to maintaining pressure control, protecting the environment, and ensuring long-term well reliability and operational efficiency.

This approach utilizes the use of distributed temperature sensing (DTS) alongside the new generations of distributed acoustic sensing (DAS) systems. This combination is used to monitor well integrity breaches and diagnose the source of sustained annulus pressure. Hybrid wireline Fiber optic cables are deployed into wells up to the maximum reachable depth to collect real-time data that are further processed to diagnose the exact source of sustained annulus pressure (SAP) and track the effluent to its source. Identifying the specific depths and the sources responsible is critical so an efficient workover plan can be made to mitigate and bring the well back under production. Based on this detailed diagnosis, targeted mitigation strategies can be developed, including focused workover operations tailored to the diagnosis. Well planned workover activities executed with precision can eliminate the root causes of pressure buildup.

The use of optical fiber-based sensing significantly enhances the ability to detect and analyze complex pressure behavior that might otherwise go unnoticed with conventional monitoring tools. Real-time DAS data allow for the identification of localized acoustic anomalies related to fluid movement. Simultaneously, DTS enables the visualization of temperature gradients that reflect dynamic thermal behavior linked to flow. The fusion of these datasets provides a complete diagnostic picture, allowing for pattern recognition across different depths and time intervals. With preplanned well manipulation tailored for each objective this innovative technology supports real time data visualization and comprehensive coverage, enabling us to understand evolving downhole conditions relating to fluid dynamics downhole. Data acquisition time using fiber optic sensing is substantially shorter than conventional methods thus improving overall efficiency. Statistically saving over 70% of the rig time consumed during conventional stations-based survey. This distributed sensing approach establishes a new standard for proactive annular pressure characterization by combining high-resolution data capture with scalable deployment and advanced interpretation workflows

This demonstrates the innovative use of fiber optic technology as a reliable diagnostic tool for source of sustained annulus pressure identification. The integration of DTS and DAS systems provided unprecedented accuracy in identifying pressure sources, facilitating targeted and efficient remediation. This approach represents a significant advancement in well integrity management, offering a new standard for diagnosing and resolving SAP.

Introduction

The challenge of managing annular pressure behavior in complex well architectures has driven the need for more precise and real-time surveillance tools. Traditional methods such as single point acoustic and temperature surveys or production logging tools offer limited spatial resolution and often rely on snapshot measurements that miss transient behaviors. Optical fiber-based sensing technologies, including Distributed Acoustic Sensing (DAS) and Distributed Temperature Sensing (DTS), have emerged as powerful tools in the surveillance landscape. Their ability to continuously monitor acoustic and thermal responses along the entire wellbore provides a novel avenue for diagnosing annular pressure dynamics, confirming isolation, and evaluating the effectiveness of remedial actions.

Recent advancements in engineered fibers and DAS interrogator architecture have improved sensitivity and reliability, enabling broader deployment. As the oil and gas industry pushes for more efficient and non-intrusive diagnostic techniques, the integration of optical fiber-based sensing into well integrity workflows is rapidly gaining traction. According to [Farhadiroushan et al. \(2024\)](#), distributed sensing technologies are transitioning from niche diagnostics to standard operational layers, particularly for annular surveillance in complex completions.

This paper proposes a workflow to monitor annular pressure behavior, perform targeted remedial actions, and verify the outcome of intervention. This underscores optical fiber-based sensing as a strategic surveillance tool—not just for detection, but for closure and confirmation across the full lifecycle of a pressure anomaly response.

Technology Overview: Optical Fiber-Based Sensing in Downhole Surveillance

Optical fiber-based sensing utilizes engineered optical fibers ([fig.1](#)) installed along the wellbore to function as dense arrays of acoustic and thermal sensors. Another critical component of any optical fiber-based sensing deployment is the interrogator unit ([fig.2](#))- a surface-based laser system that emits light pulses and interprets the backscattered signal from the fiber. These interrogators are tailored for either acoustic (DAS) or temperature (DTS) sensing and play a pivotal role in enabling high-resolution, distributed measurements.

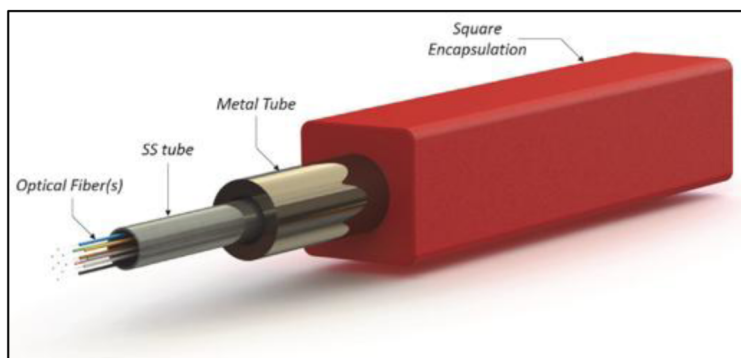


Figure 1—Fiber optic cable (Representative illustration)

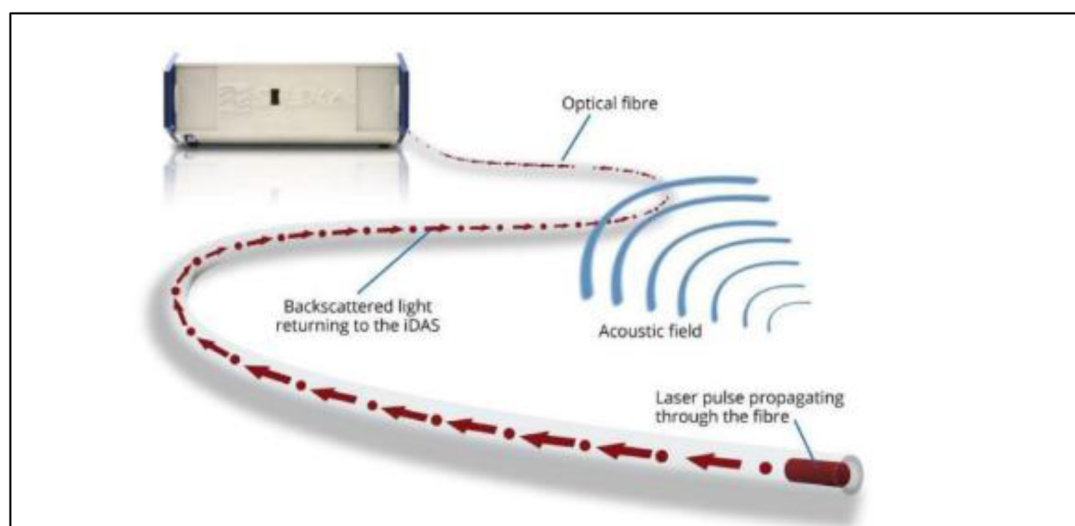


Figure 2—Basic principle of optical fiber-based sensing

DAS systems convert fiber into thousands of virtual microphones, capturing strain-rate data at meter-scale resolution. DTS measures temperature variations along the same cable with high fidelity. Combined, these sensors enable real-time, distributed data acquisition capable of revealing subtle changes in flow, temperature, or pressure dynamics.

Methodology

Two primary optical fiber-based sensing measurement techniques used in well assessment are Distributed Acoustic Sensing (DAS) and Distributed Temperature Sensing (DTS):

Distributed Acoustic Sensing (DAS)

DAS systems utilize Rayleigh backscattering of light within optical fibers. When an interrogator unit sends pulses of laser light into the fiber, tiny variations in the scattered light caused by strain (due to acoustic energy) are captured. These strain-rate signals are converted into acoustic responses. Each segment of the fiber acts as a virtual microphone.

Advanced DAS deployments now use engineered fibers with enhanced backscatter profiles, such as Constellation™ fibers, which increase the sensitivity and broaden the operational bandwidth. The engineered structure provides improved backscatters and thus in turn improves signal-to-noise ratio.

Interrogator units can be configured to record up to 2.5 kHz with spatial resolution tuned to 2 m or 10m. This allows identification of coherent acoustic energy generated by even micro-flow events. Waterfall maps and frequency spectrums were used to isolate persistent or transient anomalies along the wellbore.

Distributed Temperature Sensing (DTS)

DTS leverages Raman backscatter to measure absolute temperature along the length of the fiber. When light propagates through the fiber, a portion is backscattered, and the Stokes/anti-Stokes shift depending on the local temperature. DTS systems provide temperature resolution of $\pm 0.01^\circ\text{C}$ with a spatial resolution from 30cm.

DTS technology is fundamentally based on inelastic light scattering, specifically Raman scattering, within the fiber core. When a high-intensity laser pulse travels down the optical fiber, it interacts with the vibrational modes of the glass molecules. These interactions generate both Stokes (lower energy) and anti-Stokes (higher energy) scattered light components. The anti-Stokes component is highly sensitive to local

temperature because it arises from thermally excited vibrational states. As a result, the intensity ratio of anti-Stokes to Stokes backscatter directly correlates with the temperature of the fiber at each point.

This backscattered light travels back to the interrogator, where signal processing algorithms decode the intensity ratios and map them to spatial positions using the known light propagation speed in the fiber. Because the process is distributed, temperature readings can be extracted along the length of fiber.

The optical time-domain reflectometry (OTDR) approach used in optical- fiber based sensing systems also enables precise localization of anomalies. Using OTDR as part of QC for fiber optic cables minimizes the chances of errors or data loss, thereby improving the reliability of measurements collected during well interventions (Fig 3.).

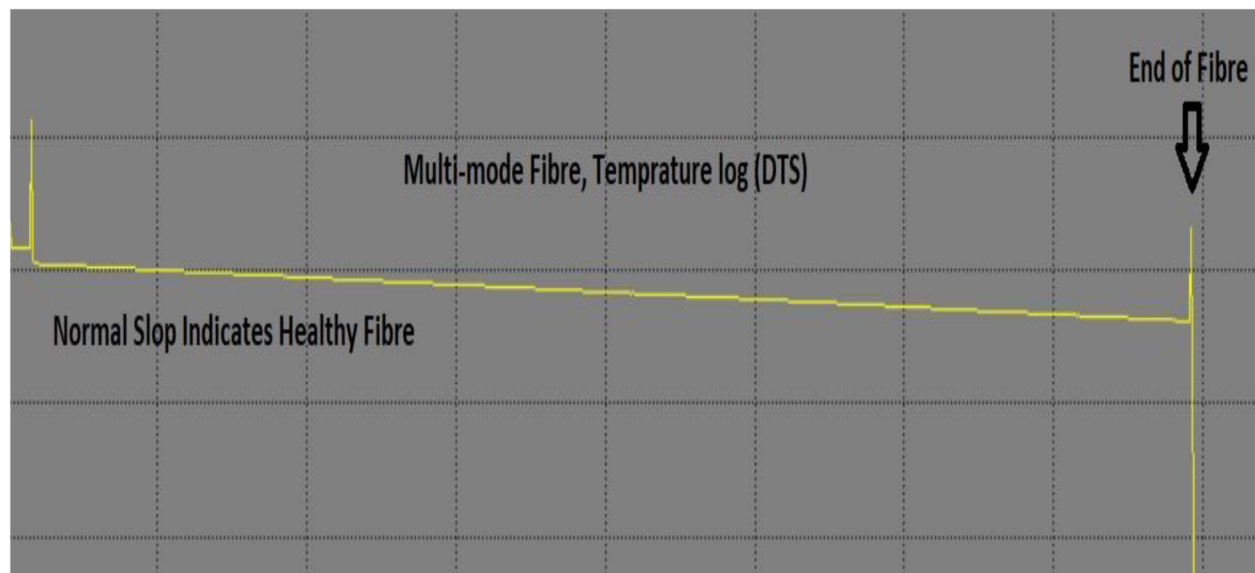


Figure 3—Example of OTDR

The DAS response is particularly sensitive to acoustic energy generated by fluid movement, friction, and flow turbulence. Enhanced signal-to-noise performance—especially with engineered fibers such as Constellation™ types—makes DAS suitable for detecting low-frequency signals associated with annular flow or communication. DTS complements this by identifying thermal anomalies due to crossflow and/or Joule-Thomson effects. Together, DAS and DTS provide dual-channel diagnostic lens.

Several examples (Jacobsen, 2024; Hudson, 2025) have shown that combining DAS/DTS analytics further enhances the ability to detect early-stage anomalies and optimize zonal isolation planning. This convergence of hardware and software continues to push optical fiber-based sensing into the mainstream of well surveillance.

Wireline Deployed Fiber optic sensing as a diagnostic and Validation method: A hypothetical case

Example (hereafter referred to as Well A, is a hypothetical well) is completed with a 13 3/8-in. surface casing, a 9 5/8-in. intermediate casing, and a 7-in. production liner. Assuming an annular pressure behavior of interest was noted in the CCA1 annulus (between the 13 3/8-in. and 9 5/8-in. casing), which demonstrated a gradual pressure increase during shut-in periods. Such a trend is indicative of unwanted fluid migration and needs a conclusive origin tracking for workover planning.

In this proposed method, a wireline-deployed optical fiber based sensing system can be employed to acquire detailed downhole measurements over the entire wellbore. The system's primary objective will be to identify and localize pressure-related anomalies along the completion. This can be achieved

through the acquisition of high-resolution distributed acoustic sensing (DAS) and/or distributed temperature sensing (DTS) data, enabling the detection of subtle variations in wellbore response that may be indicative of pressure communication, crossflow, or other dynamic events. The ability to spatially correlate these anomalies with specific completion intervals provided valuable insight into the underlying pressure pathways.

Following the initial survey and diagnosis, a targeted intervention can be executed to remediate the identified issue. A subsequent optical- fiber based sensing survey can then be conducted using the same deployment methodology to evaluate the effectiveness of the intervention. This repeat measurement process allows for direct before-and-after comparisons, significantly reducing interpretation uncertainty and confirming whether the intervention successfully mitigated the anomaly.

The deployed system can utilize retrievable optical fiber cable conveyed via standard wireline/slickline/Coil tubing operations, enabling temporary installation without the requirement for a permanently installed fiber infrastructure. This capability allows for rapid mobilization and deployment in a wide range of wells, while minimizing operational disruption and avoiding any modification to existing completion hardware. The wireline/slickline/Coil tubing-conveyed fiber provided continuous, high-resolution measurements along the full length of the well, allowing operators to detect both localized and distributed changes in wellbore conditions.

This dual-purpose methodology combining both diagnostic and post-intervention verification phases proves to be a highly effective approach for assessing annular pressure behavior and validating the success of remedial actions. Incorporating this process into future well integrity and production optimization workflows can enhance operational efficiency, reduce non-productive time, and improve decision-making by providing definitive, fiber-based evidence of treatment effectiveness (Fig 4).

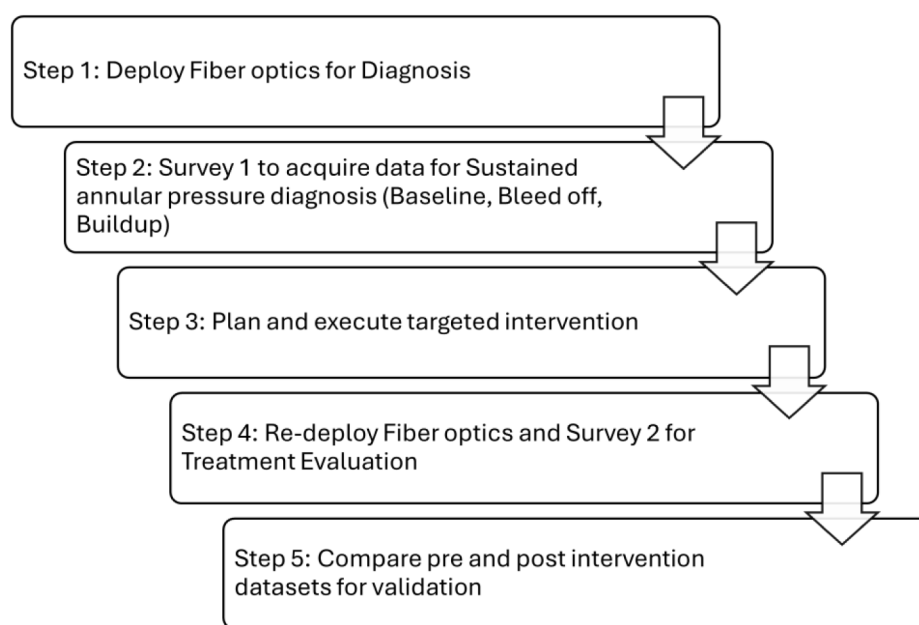


Figure 4—Proposed Workflow for diagnosis and treatment evaluation

Each survey should be planned for multiple well conditions to understand the downhole flow dynamics:

- Shut-in data acquisition for 2 hours as a benchmark for changing dynamic conditions.
- Bleed off CCA1 (or annuli under investigation) well with annuli fully open to activate the flow path relating to annuli pressure behavior for 2 hours.

- Buildup of CCA1 (or annuli under investigation) by closing the annuli to see the changes during pressure recovery.

Based on the results of survey 1 a workover can be planned and subsequent survey 2 can be conducted. This time-lapse methodology has been widely used to confirm treatment effectiveness in prior optical fiber-based sensing implementations (Farhadiroushan, 2024).

As a comparative analysis diagnosis using single point measurement for this hypothetical well (HUD at 5000 ft) is shown in table 1. below (assuming down pass speed 30 ft/min for both single point measurement and optical- fiber based sensing survey). For single point acoustic measurement 10 ft station along with 60 seconds per station is used for calculation (actual station duration could be 45 seconds, for ease of calculation 60 seconds is used, to consider for movement between stations). Please note that actual timings could vary, this is just an example for illustrative purposes.

Table 1—Time wise comparison for data acquisition

Well condition	Time for data acquisition Single point measurement (Down pass + Up pass)	Time for acquisition optical- fiber based sensing	Notes:
Baseline	Approx. 11 hours	2 hours	Optical- fiber based sensing will have one-time down pass: 2 hrs. 50 mins
Bleed off CCA1 (or annuli under investigation)	Approx. 11 hours	2 hours	
Buildup CCA1(or annuli under investigation)	Approx. 11 hours	2 hours	
Total time	Approx 33 hours	Approx 9 hours	Optical- fiber based sensing time inclusive of Run in hole time

An approximate 70% saving in data acquisition duration can be observed from the example provided above for a single annuli survey. This significant reduction in time demonstrates the remarkable efficiency of optical- fiber based SENSING technology in acquiring data in a fraction of the time compared to traditional methods. The time saved in this single instance shows the exceptional speed and operational efficiency of optical- fiber based SENSING, which allows for faster and more accurate measurements.

This level of time saving in a one-time data acquisition is crucial because it opens the possibility of repeated and more frequent assessments without a substantial increase in operational costs. Moreover, this efficiency is not just a time benefit but a critical advantage for conducting timely evaluations during intervention operations. The ability to reassess downhole conditions rapidly, with minimal time investment, is what makes optical- fiber based sensing a highly practical and cost-effective option for diagnosing and validating the success of intervention measures in real-time.

Diagnostic survey

Optical fiber-based sensing data is captured while the well is in shut-in in condition, bleed off and buildup condition as a part of investigation. Comparison of the data between dynamic condition (bleed off and buildup) with respect to baseline will support to identify the suspected zone causing pressure buildup in the annuli under investigation. Average traces (typical for DTS) and Waterfall maps (for DTS and DAS) are two methods used to visualize the data. A DTS trace is an average temperature over a chosen period to represent the temperature during one well condition of choice (Fig 5). DTS traces can be created for any well condition of choice. Another type of visualization used is waterfall maps. Waterfall plots for Distributed Acoustic Sensing (DAS) display acoustic energy intensity over time and depth, offering a visual representation of

dynamic fluid activity in the well. In these plots, time typically progresses along X axis while depth spans Y axis, allowing us to observe and track all acoustic events.

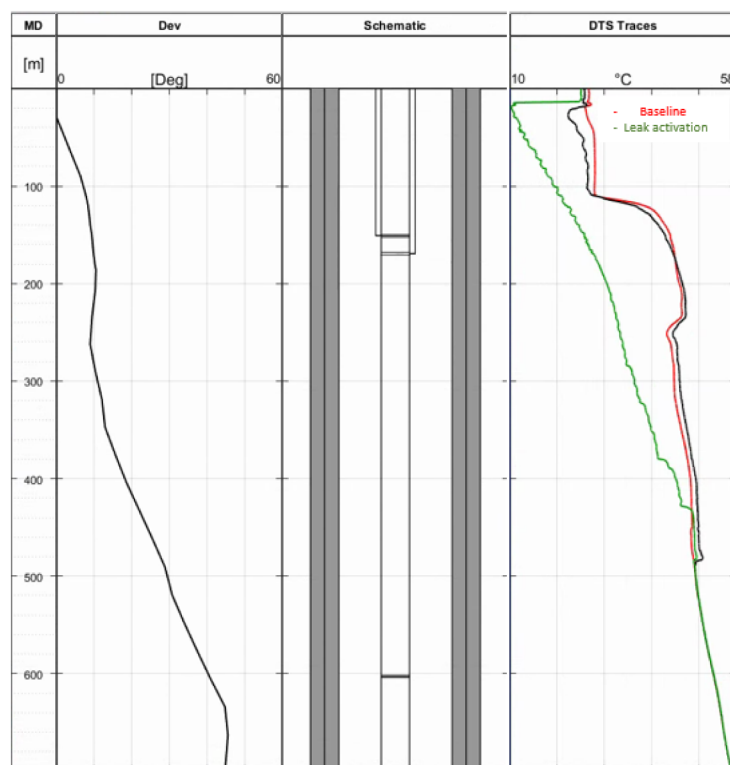


Figure 5—Example of DTS traces (Bazaid et al.,2025)

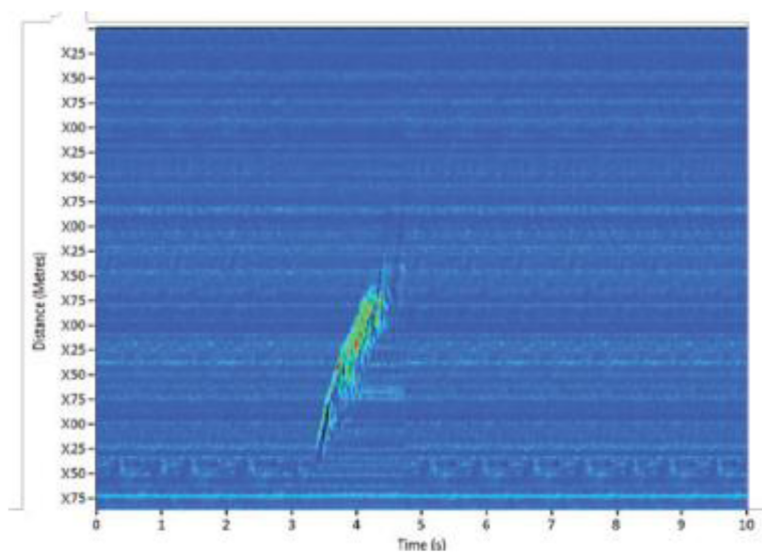


Figure 6—Leak detected using DAS (Naldrett et al., 2018)

Optical- fiber based sensing technology offers the capability to generate an acoustic spectrum for any specified duration, providing real-time, continuous monitoring of acoustic signals. By leveraging the sensitivity of fiber optic cables, optical- fiber based sensing can capture detailed acoustic data over a wide range of time intervals, from short bursts to extended durations. This flexibility allows for precise acoustic profiling, whether it is required for brief monitoring periods or long-term assessments. With optical- fiber

based sensing, operators can obtain high-resolution acoustic spectra over any chosen timeframe, enabling enhanced analysis and diagnostics of well conditions (Fig 6).

In the proposed methodology, both Distributed Temperature Sensing (DTS) and Distributed Acoustic Sensing (DAS) are utilized to investigate the source of pressure buildup within well annuli. DTS provides a continuous thermal profile along the wellbore, enabling the identification of localized temperature anomalies that may be associated with fluid movement, crossflow between zones, or changes in annular fluid composition. DAS complements this by recording acoustic energy along the same fiber, allowing detection of transient or sustained noise events that can be correlated with fluid ingress, migration, or mechanical activity. The combined interpretation of DTS and DAS datasets enables full-length well coverage in a single survey, reducing the need for multiple passes with conventional tools. This integrated approach improves the ability to localize the depth of pressure sources and characterize the associated flow mechanism, thereby supporting more informed intervention planning.

Validation Survey (Post intervention)

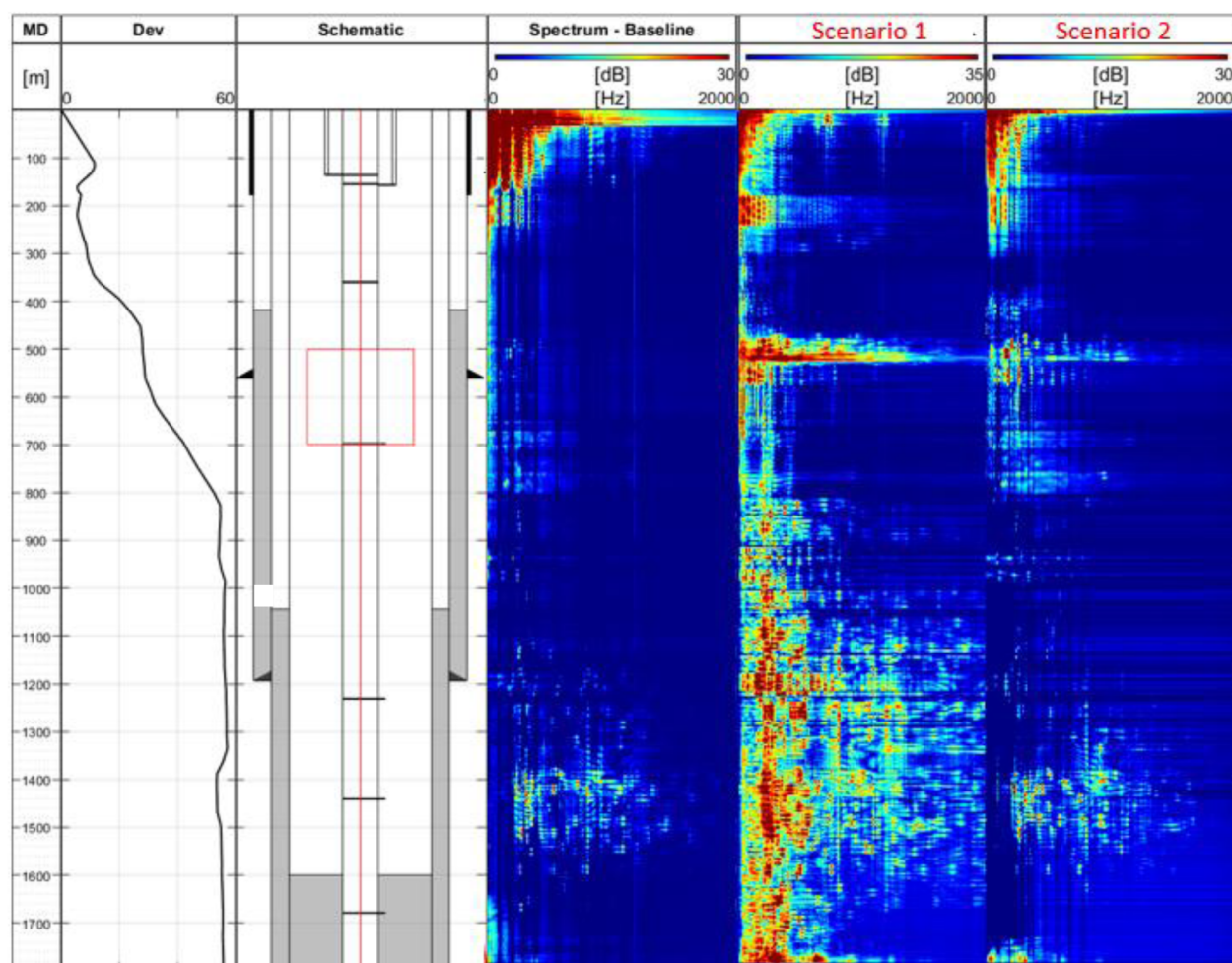


Figure 7—Visualization to show decrease in acoustic amplitudes over different conditions. (Bazaid et al.,2025)

Following the results and interpretations from the initial optical fiber-based sensing survey, a targeted workover can be performed to address the annular pressure anomaly. The treatment is aimed to restore hydraulic isolation and suppress unwanted fluid migration. Post-intervention surface pressure measurements can indicate a reduction in annular pressure buildup. During bleed-off, pressure drops swiftly from pre-job levels to atmospheric conditions within minutes, and the subsequent build-up over several

hours showing significantly lower peak values compared to pre-treatment behavior is a sign of treatment success. This immediate change in surface response served as the first validation of treatment effectiveness and justifies conducting a second fiber optic survey to confirm subsurface response across the wellbore (Fig 7).

The baseline survey is conducted prior to any operational changes and captures the existing acoustic signature of the wellbore. At this stage, the continuous depth-frequency record provides a valuable reference for identifying pre-existing downhole events such as crossflow between zones, fluid entry from behind casing, or other sources of sustained acoustic activity. Because the baseline spectrum serves as the benchmark against which all subsequent surveys are compared.

The methodology leverages Distributed Acoustic Sensing (DAS) to compare acoustic intensity before and after intervention, providing a direct indication of the downhole effect of the treatment. After the remedial intervention, an identical acquisition is repeated under the same operational conditions. The post-treatment spectrum is then directly compared with the pre-treatment dataset. Any significant reduction in acoustic intensity at the previously identified depth is interpreted as a measurable downhole effect of the intervention, indicating suppression or elimination of the source.

This comparative approach provides a clear, visual, and quantitative measure of treatment effectiveness. The use of consistent acquisition parameters ensures that observed changes are due to actual downhole conditions rather than variations in measurement setup. By capturing the entire wellbore in each pass, DAS ensures that both persistent and transient events are recorded, increasing confidence in the results.

Integrating DTS into the same workflow enables correlation of acoustic reductions with changes in thermal profiles, strengthening the interpretation. Together, these distributed measurements provide a robust, repeatable, and non-intrusive method for confirming intervention success and validating well integrity improvements.

Conclusion

This example study illustrates the transformative impact of optical fiber-based sensing technology in modern well integrity workflows. Through the integration of Distributed Acoustic Sensing (DAS), Distributed Temperature Sensing (DTS), optical fiber-based sensing provided a comprehensive diagnostic view of downhole behavior across time and space. The detection of thermal and acoustic active intervals during Run 1 enables a targeted intervention that and Run 2 for treatment evaluation enables a effective method for downhole confirmation.

The use of the said technologies and processing techniques enables workover engineers take the most accurate decisions while planning well work over programs while respecting the rig time and much clearer results.

Optical fiber-based sensing enabled rapid localization of active zones and real-time post-job confirmation. The data acquisition can be completed in less than 10 hours. In contrast, legacy logging tools relying on point measurements (e.g., PLT, single point temperature and acoustic survey) require more than 30 hours.

By combining DAS and DTS analysis, the operator was able to prioritize targeted remedial actions, reduce uncertainty, and verify job success without additional downhole interventions.

This example exemplifies how optical fiber-based sensing transforms well integrity workflows from reactive to predictive by providing high-resolution, continuous insight across the entire wellbore.

Time-lapse comparisons of DAS spectrum plots provided direct evidence of intervention efficacy. Energy reductions in affected intervals based on RMS acoustic amplitude metrics. This can be matched with thermal normalization in DTS. The overlay of two surveys illustrated and confirms a progressive decrease in flow-indicative acoustic energy and return of thermal symmetry post-treatment.

In other examples (Hudson, 2025; US10877001), similar time-lapse optical fiber-based sensing applications have shown comparable success in identifying subtle fluid migration behind casing and quantifying remediation effectiveness.

Optical fiber-based sensing proved especially powerful in reducing the decision cycle time. Traditional wireline logging tools, which require repeated rig-up/down and only provide snapshot data at discrete depths, are inherently slower and less complete. In contrast, optical fiber-based sensing enabled a complete diagnosis, targeted intervention planning, and post-job verification under a single operational shift. This rapid turnaround is invaluable, where downtime and deferred production come at significant cost. The result is not just identification of a flow path but detailed understanding of its initiation, directionality, and intensity. These capabilities open new pathways for zonal isolation validation, cement monitoring, and early anomaly detection.

One of the key operational challenges in DAS and DTS applications is the management of large data volumes generated during high-resolution acquisitions. Continuous recording over extended durations can produce terabytes of raw data, creating bottlenecks in storage, transfer, and post-processing workflows. An area for future improvement is the integration of AI-assisted data triage systems that can process data streams in real time, identify segments containing potential anomalies/ relevant baseline data, and prioritize them for storage and analysis. Furthermore, AI-driven compression algorithms could be applied to reduce file sizes without degrading the signal fidelity needed for accurate interpretation, thereby optimizing both storage and transmission efficiency. In parallel, machine learning models trained on historical datasets could be used to recognize characteristic acoustic or thermal signatures, enabling automated event tagging and rapid decision support. These combined capabilities would not only streamline data handling but also shorten interpretation cycles, ultimately supporting faster and more informed operational responses in dynamic well conditions.

From a broader operational perspective, this example validates optical fiber-based sensing as a viable and cost-effective tool for lifecycle integrity management. The real-time feedback loop it creates—detect, intervene, verify—significantly shortens the diagnosis-to-decision timeline. In highly instrumented or critical wells, optical fiber-based sensing should no longer be seen as an optional enhancement, but rather as a fundamental component of well architecture.

In conclusion, this example affirms the strategic value of optical fiber-based sensing for enabling faster decisions, more precise interventions, and enhanced lifecycle oversight. By replacing uncertainty with insight, optical fiber-based sensing elevates the standard of care in well surveillance to a truly data-driven discipline.

Looking ahead, the role of optical fiber-based sensing is poised to grow as operators seek more predictive, data-rich strategies for asset integrity. With also a possibility for permanent installations, now standard in many completions, the infrastructure is in place to enable continuous monitoring, time-lapse comparisons, and machine learning-driven interpretations. This example reinforces that optical fiber-based sensing is not just a diagnostic tool, but a cornerstone of modern, intelligent well management systems. optical fiber-based sensing offers significant opportunities in developing predictive models for annular pressure behavior, validating zonal isolation, and optimizing well completion strategies. The insights from this well reinforce that optical fiber-based sensing is not merely a diagnostic tool, but a continuous surveillance framework that enables data-driven development.

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